Huawei Access Node Hardware/Environmental Fault Isolation and Repair SOP KR-ACC-15 KR Puram Access

Node: KR-ACC-15 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_8825

Description

The occurrence of Hardware/Environmental Faults (ASIC memory parity errors, line card crashes, power supply redundancy loss) on Huawei Access Node KR-ACC-15 at KR Puram Access can be attributed to a complex interplay of underlying protocol and hardware mechanics. In the context of Huawei's Access Node architecture, the Line Card (LC) functions as a critical component, responsible for handling the data plane operations, including packet forwarding and filtering. The LC's performance is heavily reliant on the Advanced Switching Interconnect (ASIC), which is responsible for processing high-speed packet switching and routing. However, the ASIC's memory parity errors can lead to a cascade of failures, including line card crashes and ultimately, power supply redundancy loss. Furthermore, environmental factors such as excessive temperature, humidity, or power fluctuations can also contribute to hardware failure. Therefore, it is essential to conduct a thorough investigation and repair to isolate and rectify the root cause of the failure.

Pre-requisite Checks

- Verify the site location and confirm the Node ID as KR-ACC-15 at KR Puram Access
- Check the redundant link status to ensure that it is functioning properly
- Dispatch remote hands to KR Puram Access to perform on-site checks and maintenance

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: display fault, Expected Output: Fault information, including alarm codes and descriptions, Description: This command displays the current fault information, including the alarm code and description, which can help identify the root cause of the failure.

Command: display fault-history, Expected Output: Fault history, including previous alarm codes and descriptions, Description: This command displays the fault history, including previous alarm codes and descriptions, which can provide valuable insights into the failure
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pattern and help isolate the root cause.

Command: display board-info, Expected Output: Board information, including temperature, voltage, and current readings, Description: This command displays the board information, including temperature, voltage, and current readings, which can help identify any environmental or hardware-related issues that may be contributing to the failure.
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Resolution Steps

Step 1: Step: Verify the alarm code and description, Command: display alarm, Description: Verify the alarm code and description to confirm that it matches the reported failure and to identify the root cause.

Step 2: Step: Replace the faulty Line Card, Command: replace board , Description: Replace the faulty Line Card with a new one to rectify the hardware failure and prevent further cascade of failures.

Step 3: Step: Verify power supply redundancy, Command: display psu-info, Description: Verify power supply redundancy to ensure that it is functioning properly and that the failure is not related to power supply issues.

Post-Check Verification

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Command: display fault, Expected Output: No faults reported, Description: Verify that no faults are reported and that the node is functioning properly.

Command: display board-info, Expected Output: Normal board information, Description: Verify that the board information is normal and that there are no environmental or hardware-related issues.
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Roll-back Procedures

- Step: Rollback the configuration, Command: undo replace board , Description: If the fix fails, rollback the configuration to the previous state and restore the original Line Card.

- Step: Verify the node functionality, Command: display node-info, Description: Verify that the node is functioning properly and that the failure has been resolved.